

An aerial night view of a city, likely Shanghai, with a network overlay of glowing nodes and connecting lines. The city lights are visible, and the network lines are white and curved, connecting various points across the city. The background is a gradient of blue and white, with a large white curved shape on the right side.

# Wireless Networks

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# Wireless Networks

## Lecture-3



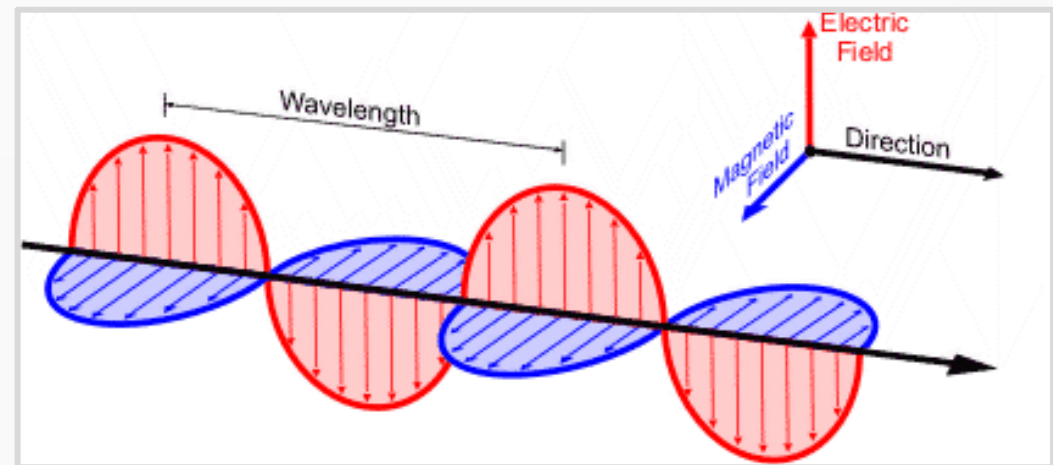
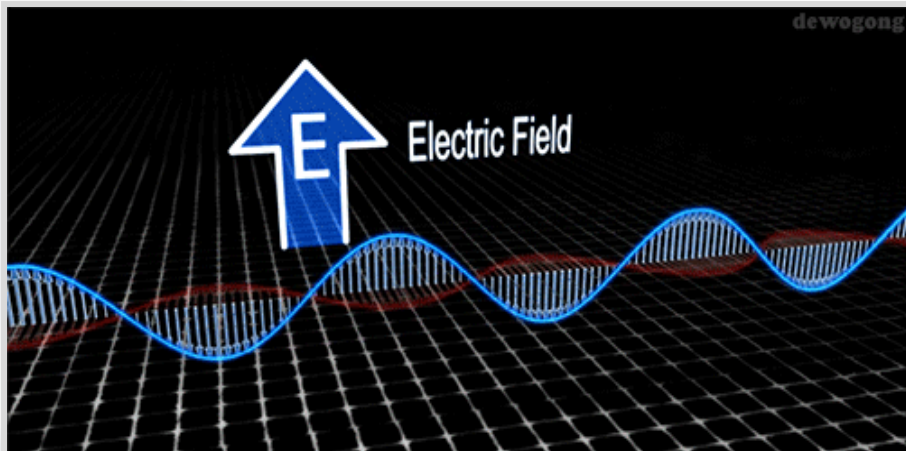
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- ✓ Electromagnetic Waves
- ✓ Propagation Types
- ✓ Antenna Idea
- ✓ What is the antenna is ?
- ✓ Antenna and Wavelength
- ✓ Antenna Types
- ✓ Basic Antenna Parameters
- ✓ Antenna Applications

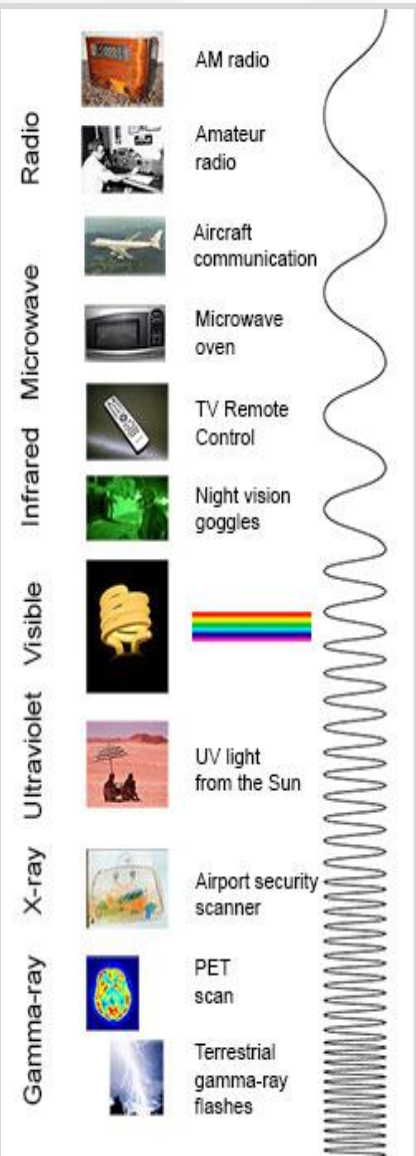


# Electromagnetic Waves

- ✓ Electromagnetic waves are **disturbances** in **electric** and **magnetic** fields that carry energy and information through space.
- ✓ A changing electric field produces a magnetic field, and vice versa .
- ✓ These fields are perpendicular to each other and to the direction of wave propagation
- ✓ Once sinusoidal fields are created they can propagate on their own.
- ✓ These waves travel at the speed of light, approximately  $3 \times 10^8$  m/s.
- ✓ Their frequencies range from radio waves to gamma rays, covering a broad spectrum.



# Electromagnetic Waves

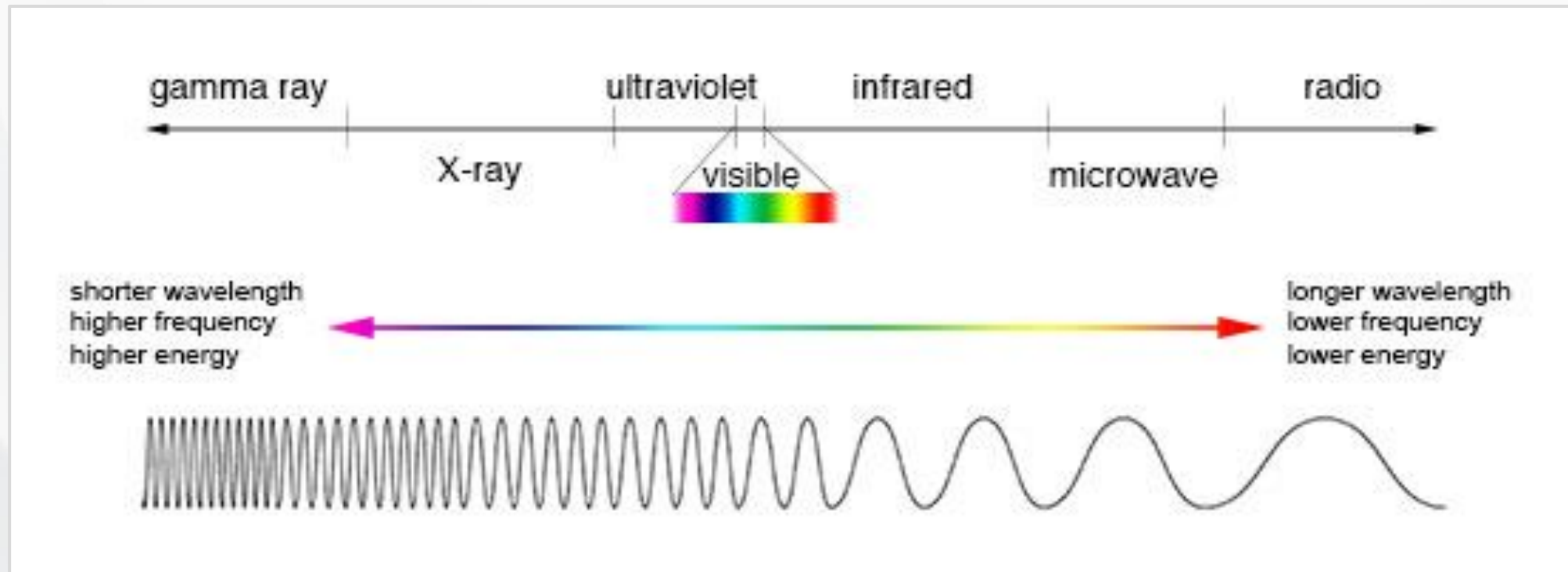


- ☑ **Radio:** Used in radio transmission and also emitted by stars and gases in space.
- ☑ **Microwave:** Like Microwave radiation in Oven.
- ☑ **Infrared:** Remote Controls, Thermal Imaging, Night Vision and Communication.
- ☑ **Visible:** Fireflies, light bulbs, and stars all emit visible light.
- ☑ **Ultraviolet:** Emitted by the Sun and are the reason skin tans and burns..
- ☑ **X-ray:** A dentist to image your teeth, and airport security and so on.
- ☑ **Gamma ray:** Doctors, fore example, use gamma-ray imaging to see inside your body.

# Electromagnetic Waves

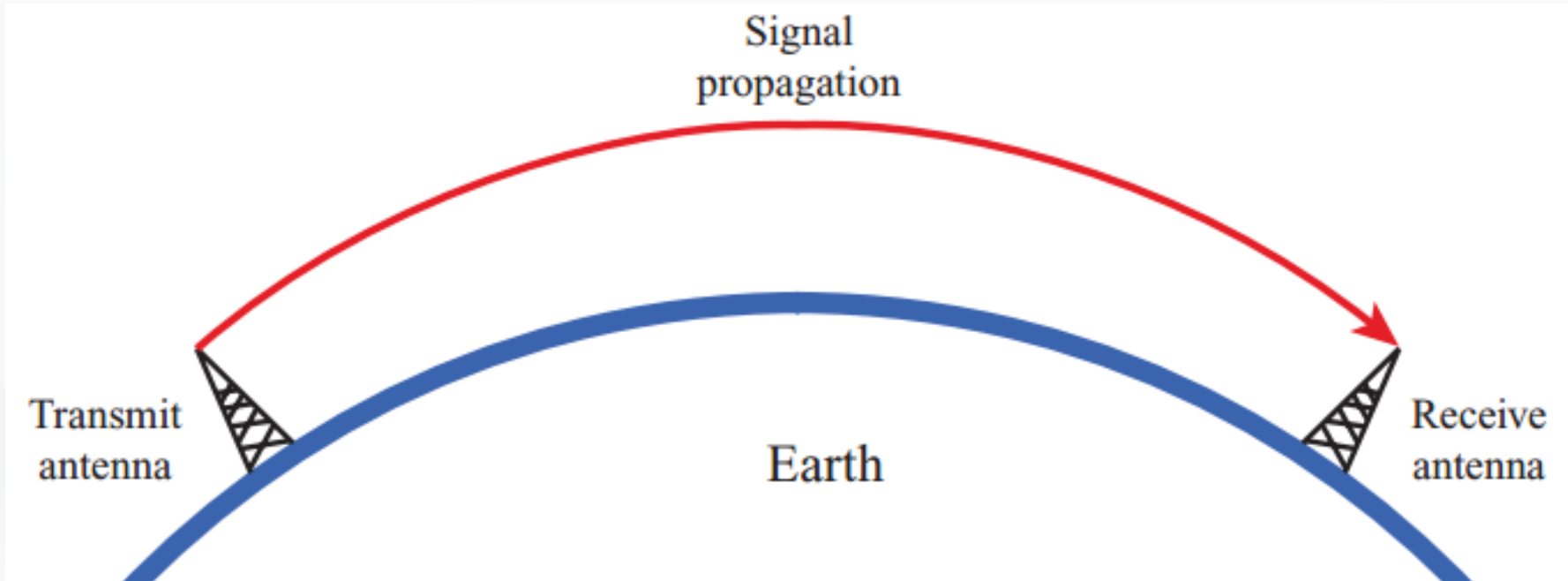
**Electromagnetic** radiation can be described by energy, wavelength, or frequency, which are **mathematically** related.

$$\lambda = \frac{c}{f} \quad E = hf \quad E = \frac{hc}{\lambda} \quad h \text{ is Planck's constant } (6.626 \times 10^{-34} \text{ J}\cdot\text{s}),$$



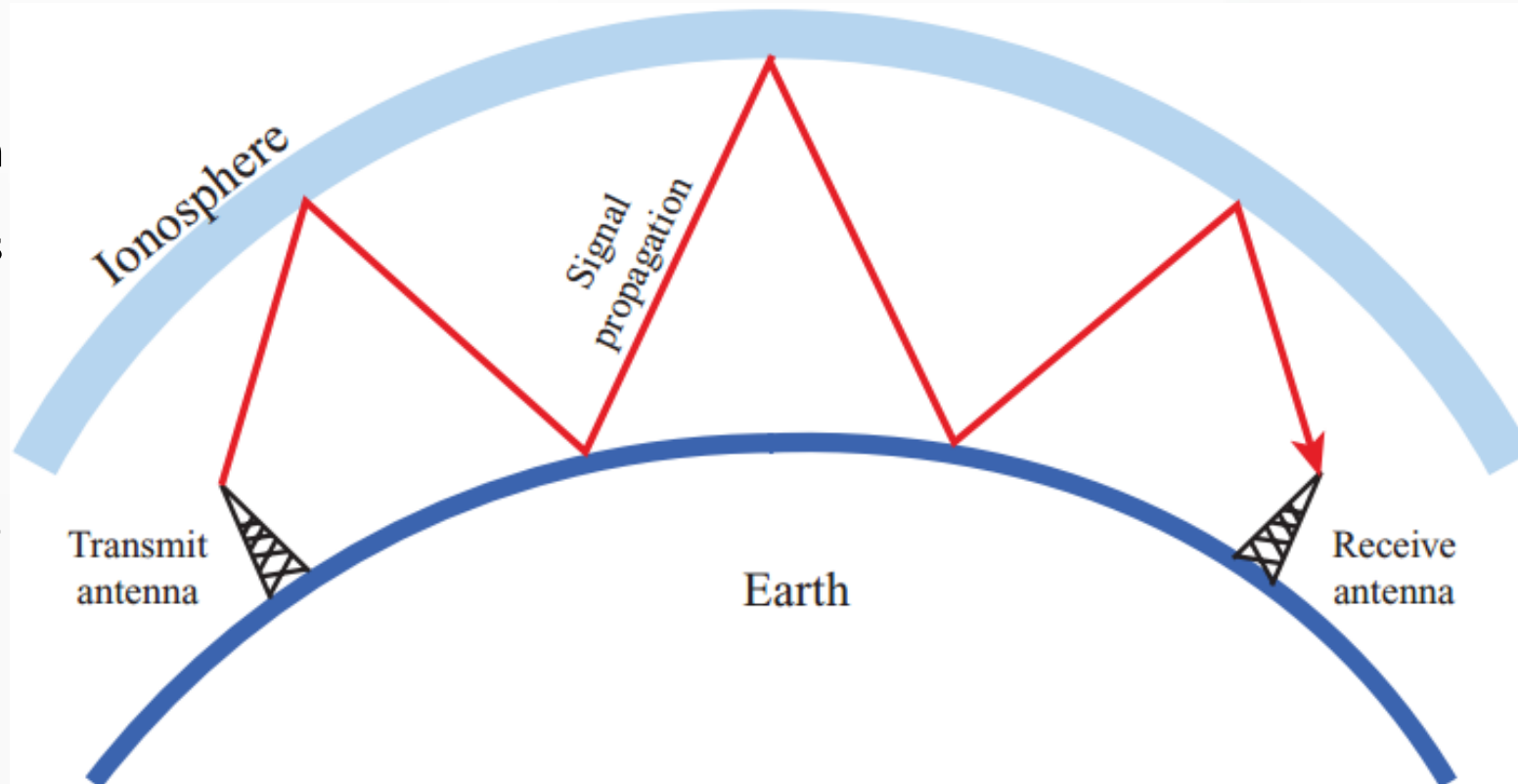
# Electromagnetic Wave Propagation Modes

- ☑ **Ground wave propagation** allows radio waves to travel along the Earth's surface, following its curvature. It is effective at low and medium frequencies, enabling communication beyond the visual horizon. This method is useful for maritime and AM radio communications, as it helps maintain contact over long distances by bending around obstacles.



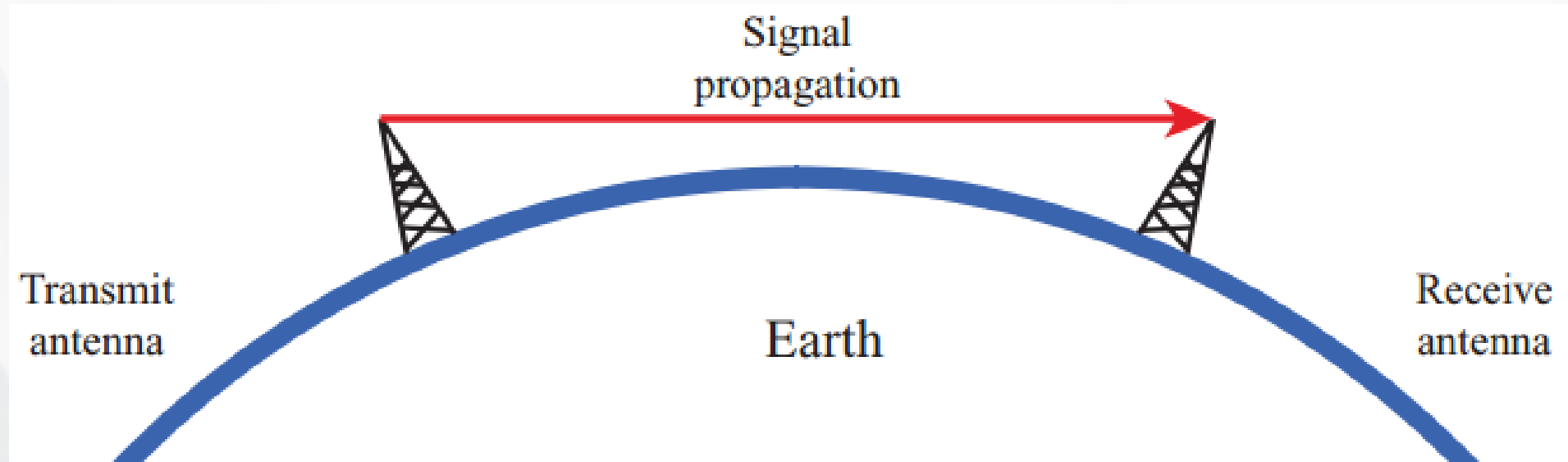
# Electromagnetic Wave Propagation Modes

- ☑ **Sky wave propagation** : a signal from an earth- based antenna is refracted from the ionized layer of the upper atmosphere (ionosphere) back down to earth.
- ☑ A sky- wave signal can travel through a number of hops, bouncing back and forth between the ionosphere and the earth's surface.
- ☑ With this propagation mode, a signal can be picked up thousands of kilometres from the transmitter.
- ☑ Sky waves generally operate between 3 and 30 MHz.



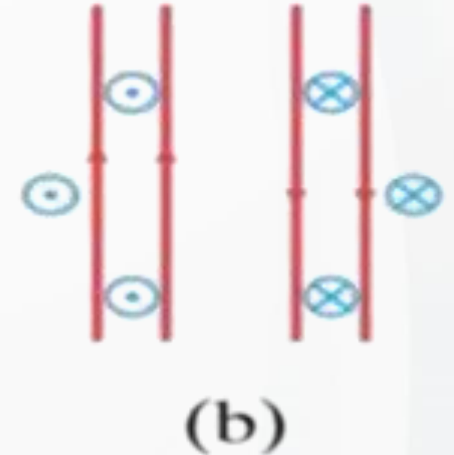
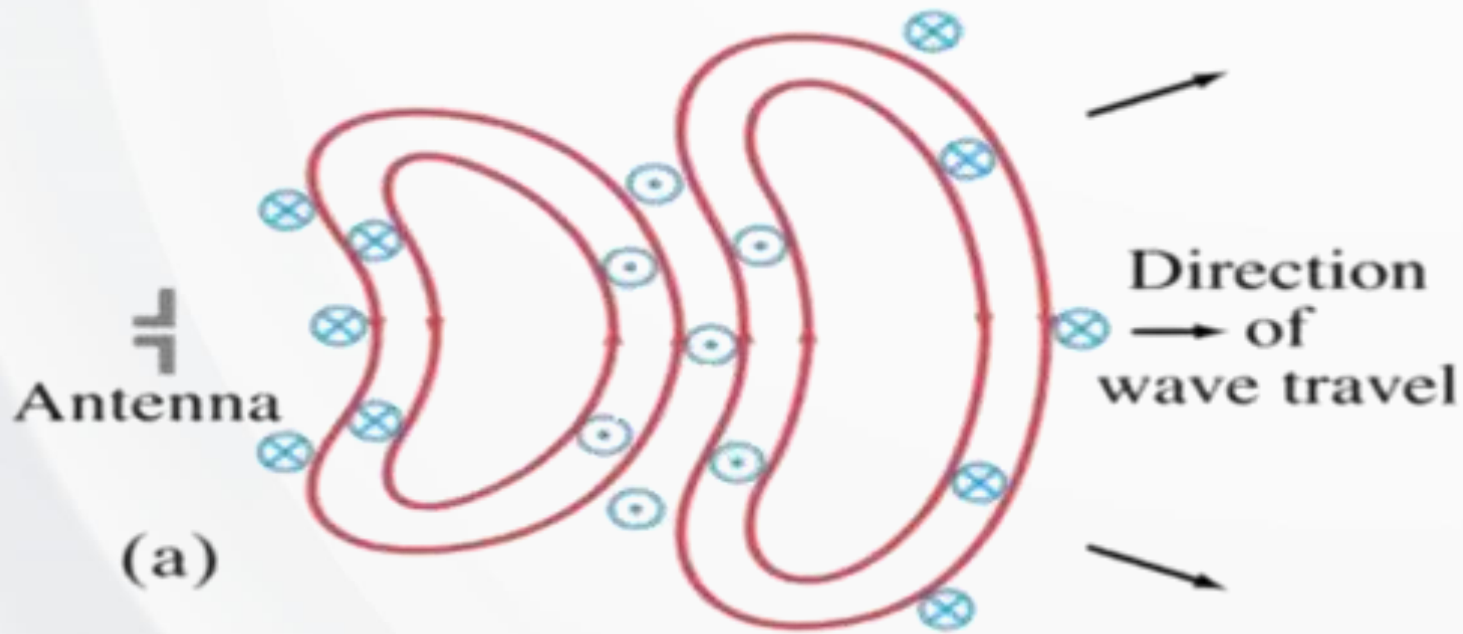
# Electromagnetic Wave Propagation Modes

- ☑ **Line- of- sight propagation (LOS):** is necessary when neither ground wave nor sky wave propagation modes can operate. This generally occurs above 30 MHz. This does not mean that line of sight always requires complete free space between transmitters and receivers, however. Different frequencies will be attenuated by atmospheric effects or have capabilities for penetrating through surfaces (e.g., through walls, buildings, cars, etc.)



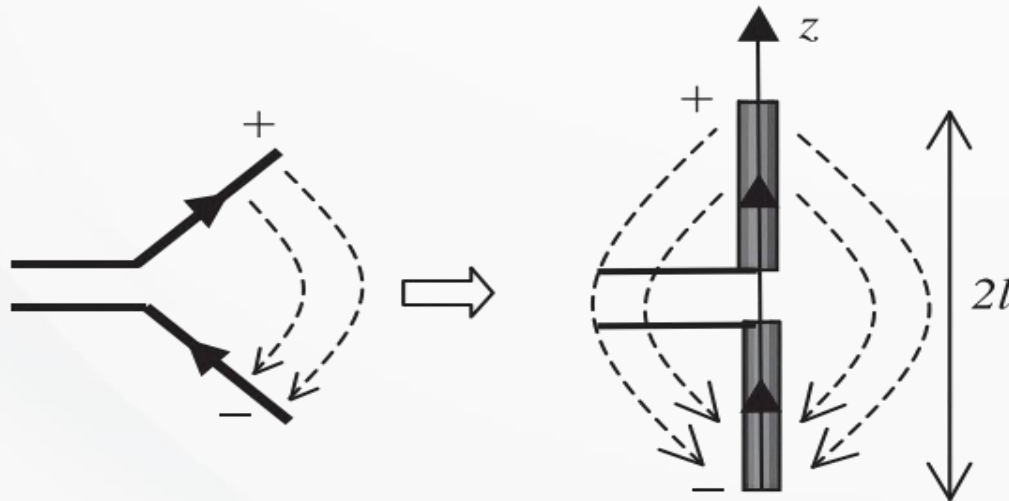
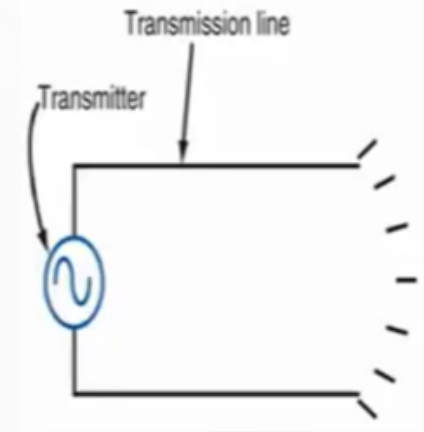
# Electromagnetic Waves: The Carriers of Information

- Electromagnetic waves are generated by oscillating electric charges.
  - The waves radiated from the oscillating charges can be detected at great distances.
- Far from the source, the waves are **plane** waves



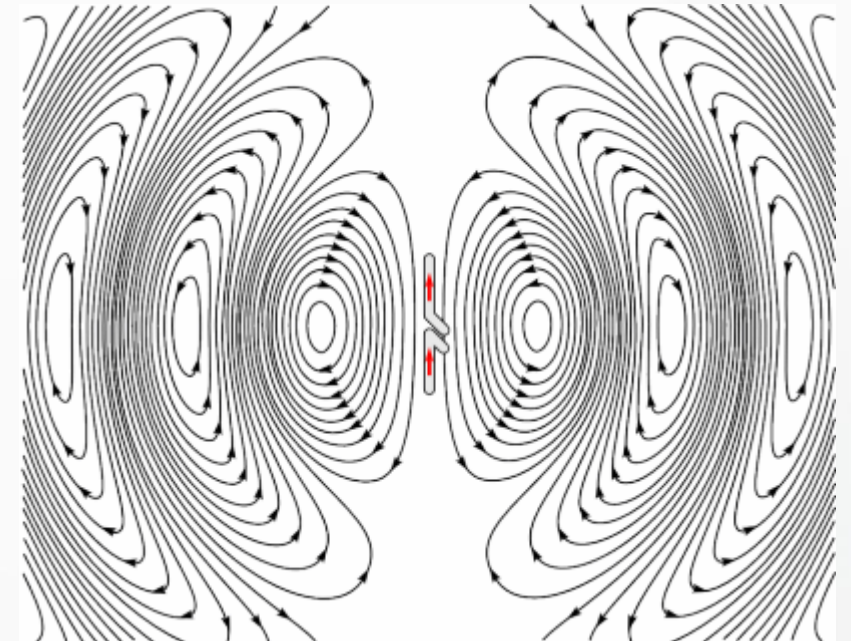
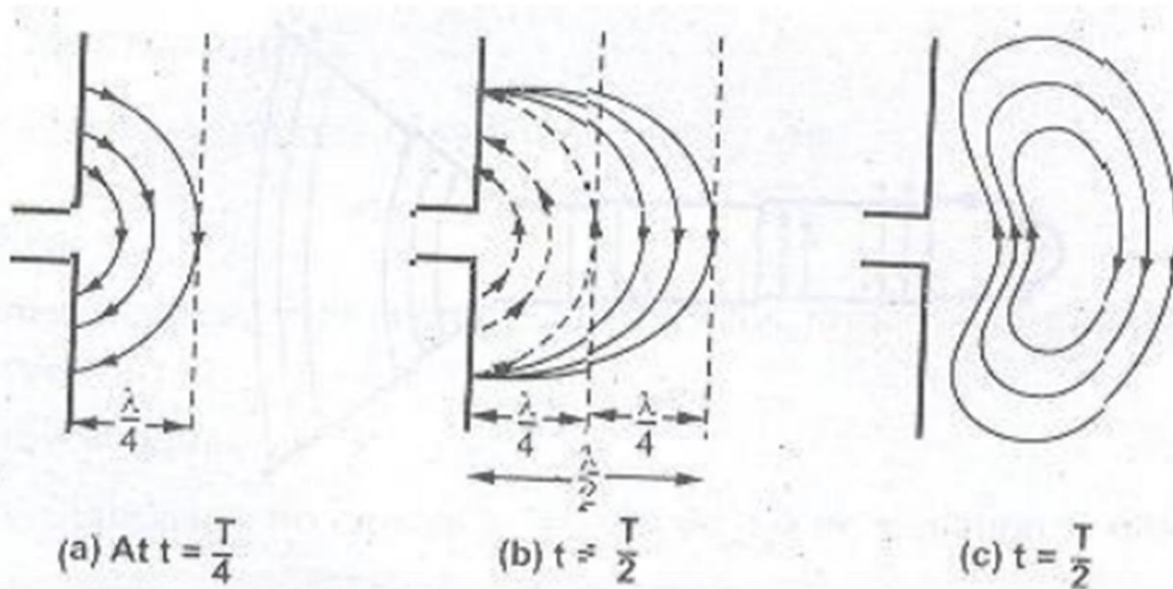
# Antenna Idea

- If a parallel-wire transmission line is left open, the electric and magnetic fields escape from the end of the line and radiate into space.
  - This radiation is **inefficient** and **unsuitable** for reliable transmission or reception.
- The radiation from a transmission line can be greatly improved by **bending** the transmission-line conductors so they are at a right angle to the transmission line.



# Antenna Idea

- The radiation from a transmission line can be greatly improved by **bending** the transmission-line conductors so they are at a **right angle** to the transmission line.



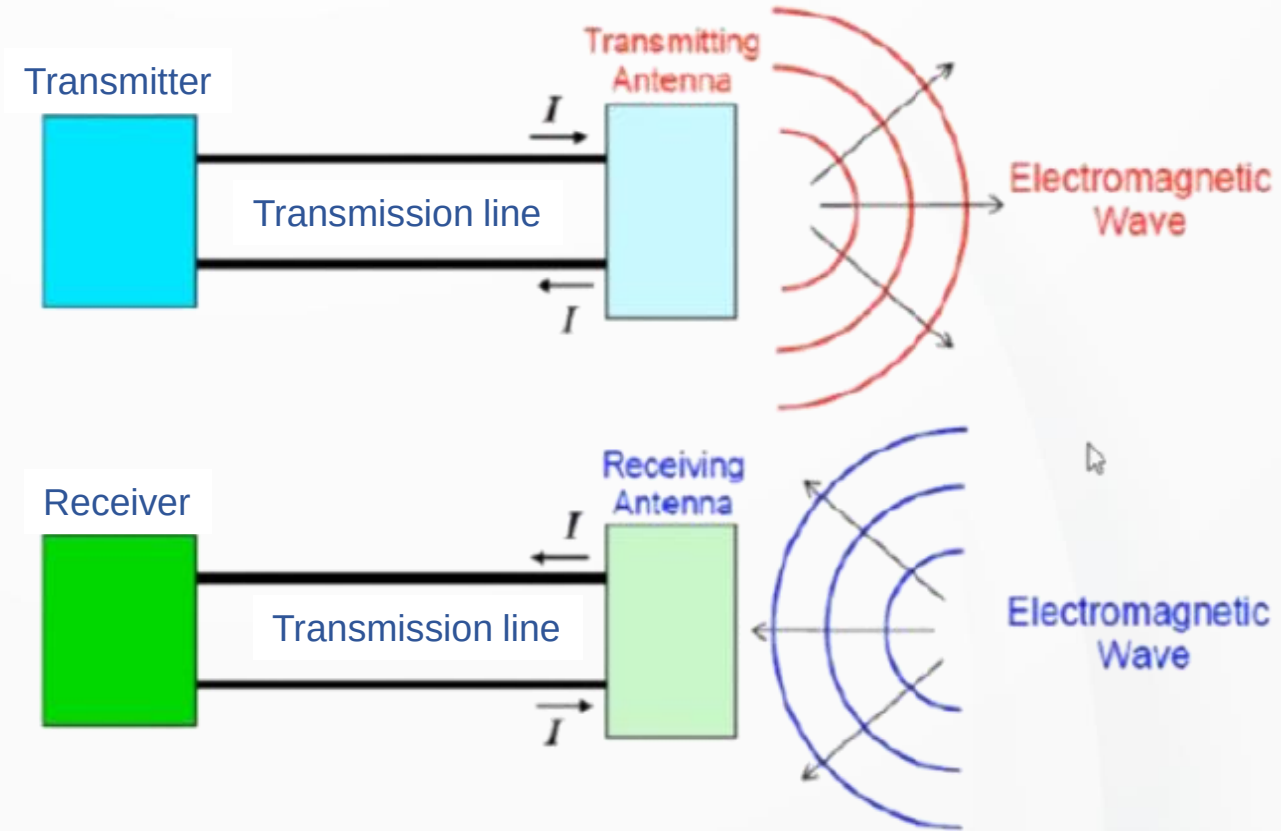
# What is the antenna is ?

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- An antenna is a usually **metallic** device used for **radiating** or **receiving** radio waves.
- An antenna is basically a **transducer**. It converts radio frequency (RF) signal into an electromagnetic (EM) wave of the same frequency and collecting EM energy.

# What is the antenna is ?

- **Transmitter** converts electrical energy into electromagnetic waves.
- **Receiver** converts electromagnetic waves back into electrical energy



**In two-way communication, the same antenna can be used for transmission and reception.**

# Antenna and Wavelength

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- Wavelength and frequency of an electromagnetic wave in free space are related by the speed of light ( $c$ )

$$c = \lambda f$$

- The length of an antenna,  $L$ , is often designed to be proportional to the wavelength ( $\lambda$ ) of the signal it is transmitting or receiving.
  - Low frequencies imply long wavelengths, hence antennas are very large.
  - High frequencies imply short wavelengths, hence antennas are usually small.
- Commonly used antenna lengths are:
  - Half-Wavelength Antenna:  $L=\lambda/2$ , Quarter-Wavelength Antenna:  $L=\lambda/4$
  - Full-Wavelength Antenna:  $L=\lambda$

# Antenna and Wavelength

## ■ Numerical Examples

### 1. Example 1: Wi-Fi (2.4 GHz)

- Frequency ( $f$ ) = 2.4 GHz =  $2.4 \times 10^9$  Hz
- Wavelength ( $\lambda$ ):

$$\lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m} = 12.5 \text{ cm}$$

- Common antenna sizes:
  - Half-Wavelength Antenna:  $\frac{0.125}{2} = 6.25 \text{ cm}$
  - Quarter-Wavelength Antenna:  $\frac{0.125}{4} = 3.125 \text{ cm}$

# Antenna and Wavelength

## ■ Numerical Examples

### 2. Example 2: FM Radio (100 MHz)

- Frequency ( $f$ ) = 100 MHz =  $1 \times 10^8$  Hz
- Wavelength ( $\lambda$ ):

$$\lambda = \frac{3 \times 10^8}{1 \times 10^8} = 3 \text{ m}$$

- Common antenna sizes:
  - Half-Wavelength Antenna:  $\frac{3}{2} = 1.5 \text{ m}$
  - Quarter-Wavelength Antenna:  $\frac{3}{4} = 0.75 \text{ m}$

# Antenna and Wavelength

## ■ Numerical Examples

### 3. **Example 3:** GPS (1.575 GHz)

- Frequency ( $f$ ) = 1.575 GHz =  $1.575 \times 10^9$  Hz
- Wavelength ( $\lambda$ ):

$$\lambda = \frac{3 \times 10^8}{1.575 \times 10^9} \approx 0.19 \text{ m} = 19 \text{ cm}$$

- Common antenna sizes:
  - Half-Wavelength Antenna:  $\frac{0.19}{2} = 9.5 \text{ cm}$
  - Quarter-Wavelength Antenna:  $\frac{0.19}{4} = 4.75 \text{ cm}$

# Antenna and Wavelength

- Numerical Examples

## **Example: AM Radio (Frequency = 1 MHz)**

1. Frequency ( $f$ ):

$$f = 1 \text{ MHz} = 1 \times 10^6 \text{ Hz}$$

2. Wavelength ( $\lambda$ ):

Using the formula:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m}$$

# Antenna Types

One type of antenna cannot fit every application. Instead, antennas come in different sizes, shapes, gain value and intended purpose.



A hospital helicopters land



A large expanse of an outdoor park.

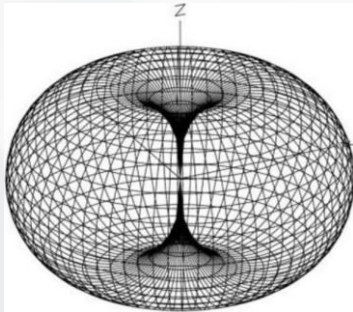


City streets where public safety vehicles travel

# Antenna Types

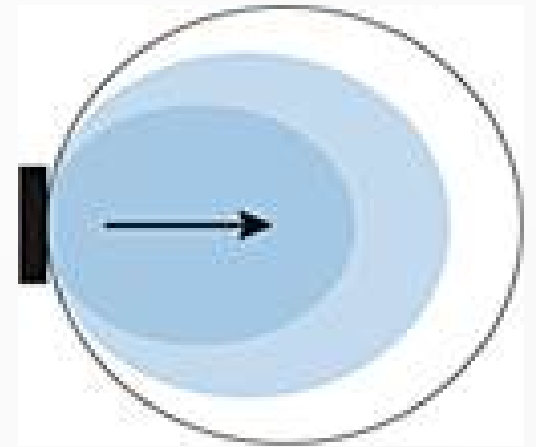
## Omni (Wire antennas)

- Whip (Monopole)
- Dipole
- Folded Dipole
- Loop
- Discone



## Directive antenna

- Array (Yagi)
- Helical
- Reflector (Parabolic)
- Horn (Aperture)
- Cell site
- Microstrip



# Antenna Types

## ■ Omni-Directional (Wire) Antenna

- Broadcasts in all directions
- Antennas are said to be **omni directional** when the power is distributed equally around the antenna without being concentrated within a certain Beam



Loop



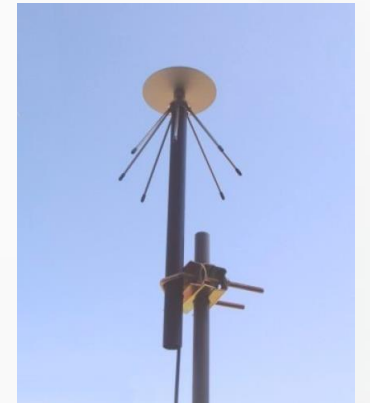
Monopole (whip)



Dipole



Folded dipole



Discone

# Antenna Types

## Antenna Types

### ■ Directional Antenna

#### □ Broadcast in a single direction

- Array antennas (Yagi)
- Helical antenna
- Reflector antennas
- Horn antenna
- Cell site antennas
- Microstrip Patch Antenna

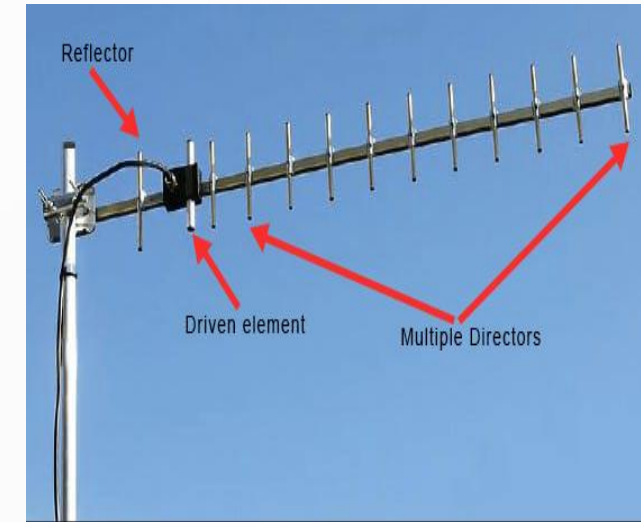
Helical



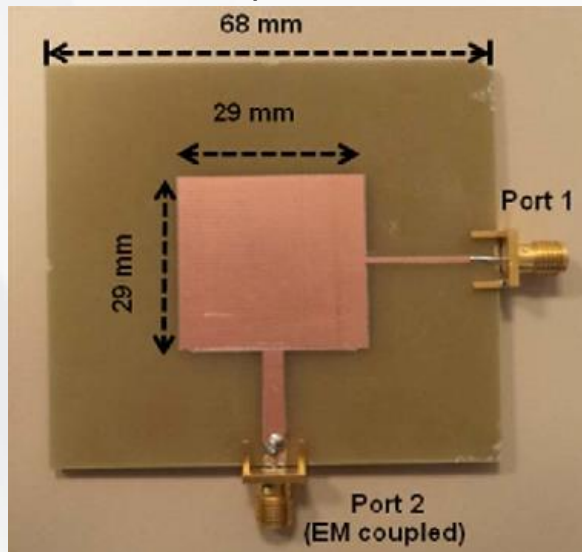
Cell site



Array antennas (Yagi)



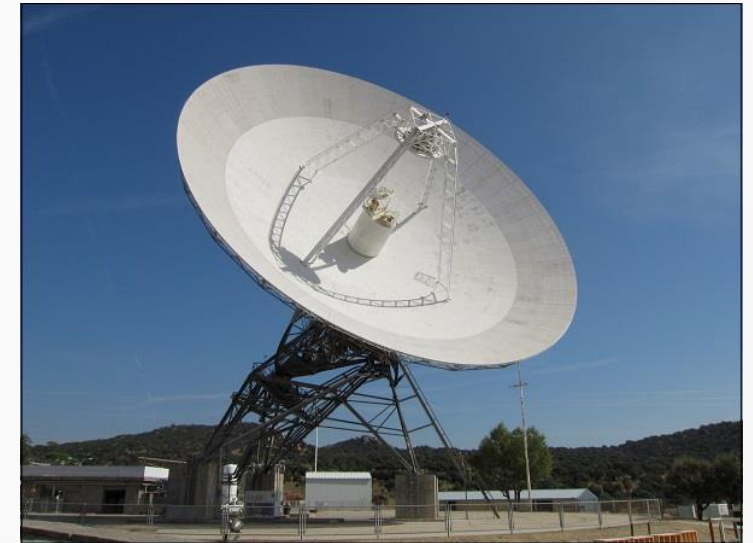
Microstrip Patch Antenna



Horn antenna



Reflector antennas



# Basic Antenna Parameters

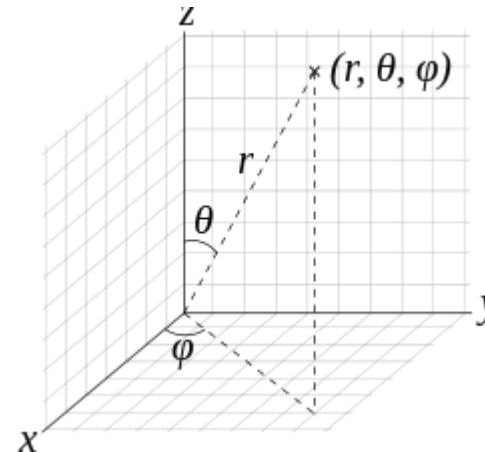
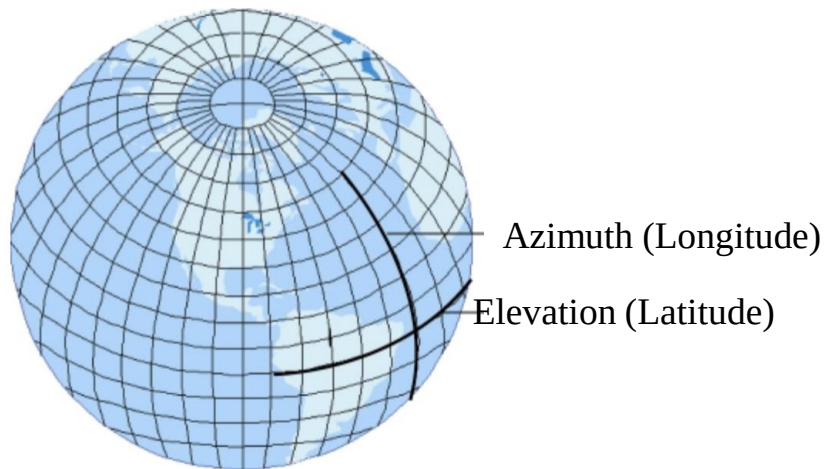
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- Radiation pattern
- Gain
- Directivity
- Radiation Resistance
- Polarization
- Efficiency
- Bandwidth
- Beamwidth

# Basic Antenna Parameters

## Isotropic Antenna

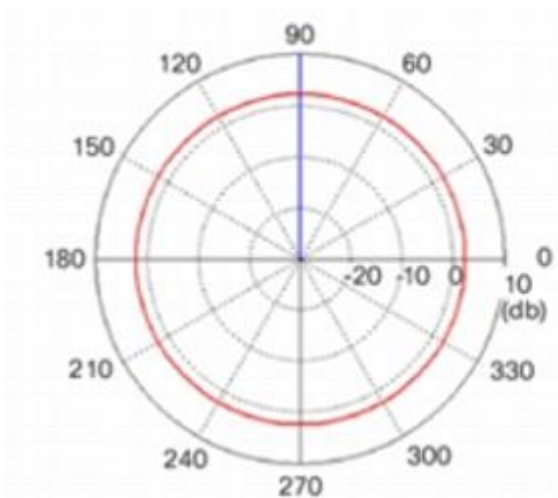
- Let's start by looking at the radiation pattern of an isotropic antenna.
- Power from an isotropic antenna is equally
- Distributed in all directions (i.e. Radiates equally in all directions)
- Antenna only exists theoretically
- Used as a useful reference to describe real antennas



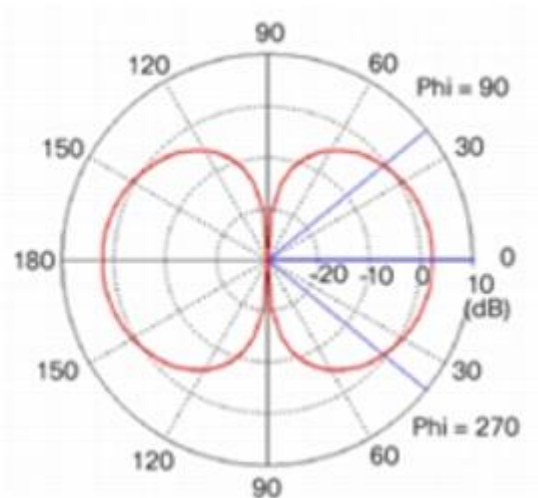
# Basic Antenna Parameters

## Radiation pattern (Omni-directional )

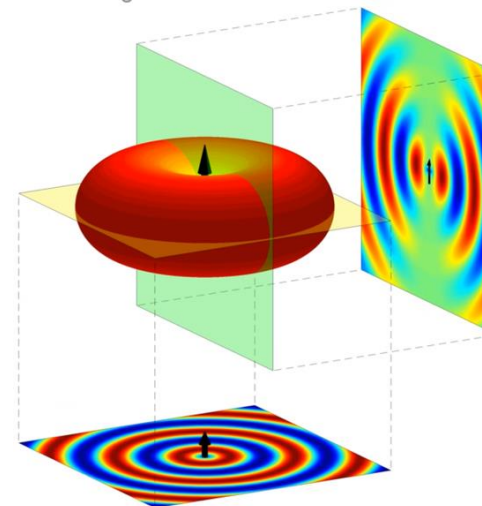
- Radiation pattern: The relative distribution of radiated power as a function of direction in space.
- Radiation pattern measurements are relative to the isotropic antenna



Dipole Azimuth Plane Pattern  
(Magnetic Field)



Dipole Elevation Plane Pattern  
(Electrical Field)



Dipole 3D Radiation Pattern



Dipole Antenna Model

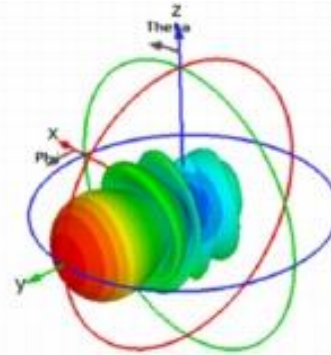
# Basic Antenna Parameters

## Radiation pattern (Directional )

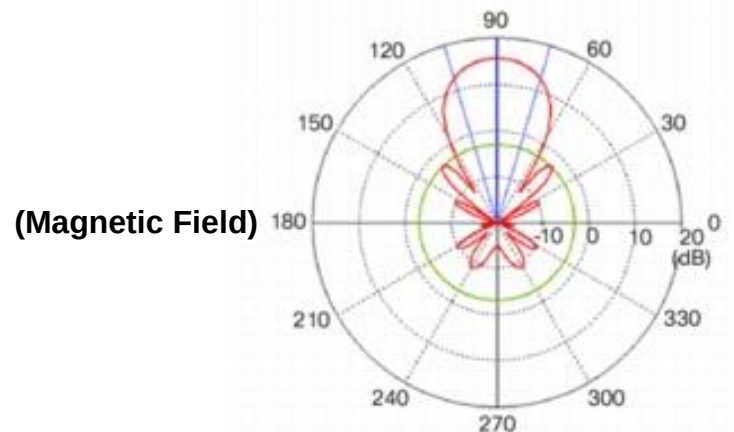
- Radiation pattern: The relative distribution of radiated power as a **function of direction** in space
- Radiation pattern measurements are relative to the isotropic antenna



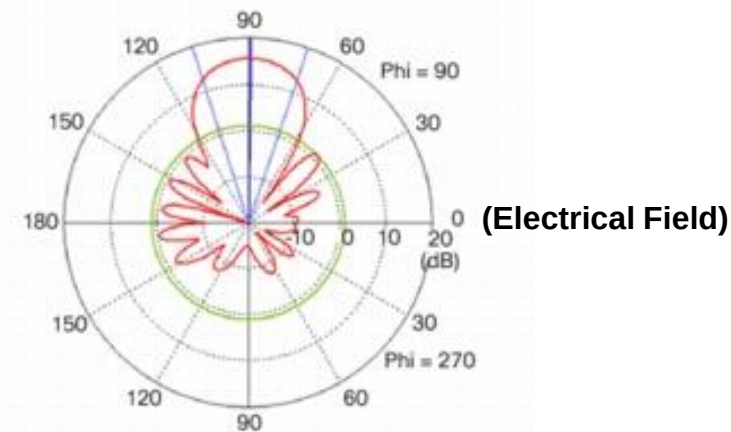
(a) Yagi Antenna Model



(b) Yagi Antenna 3D Radiation Pattern



(c) Yagi Antenna Azimuth Plane Pattern

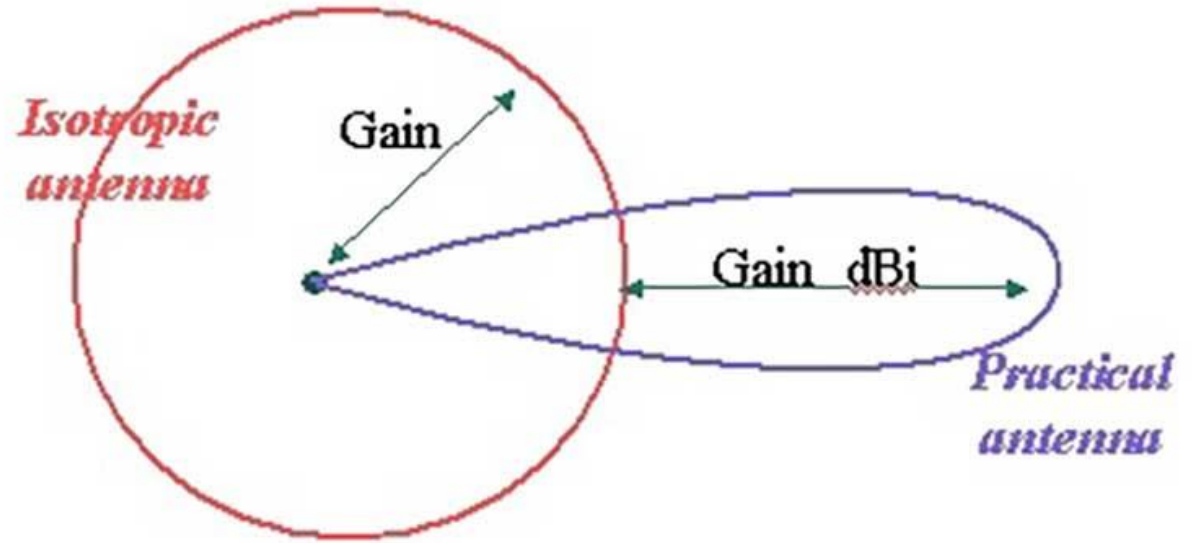


(d) Yagi Antenna Elevation Plane Pattern

# Basic Antenna Parameters

## Directive Gain

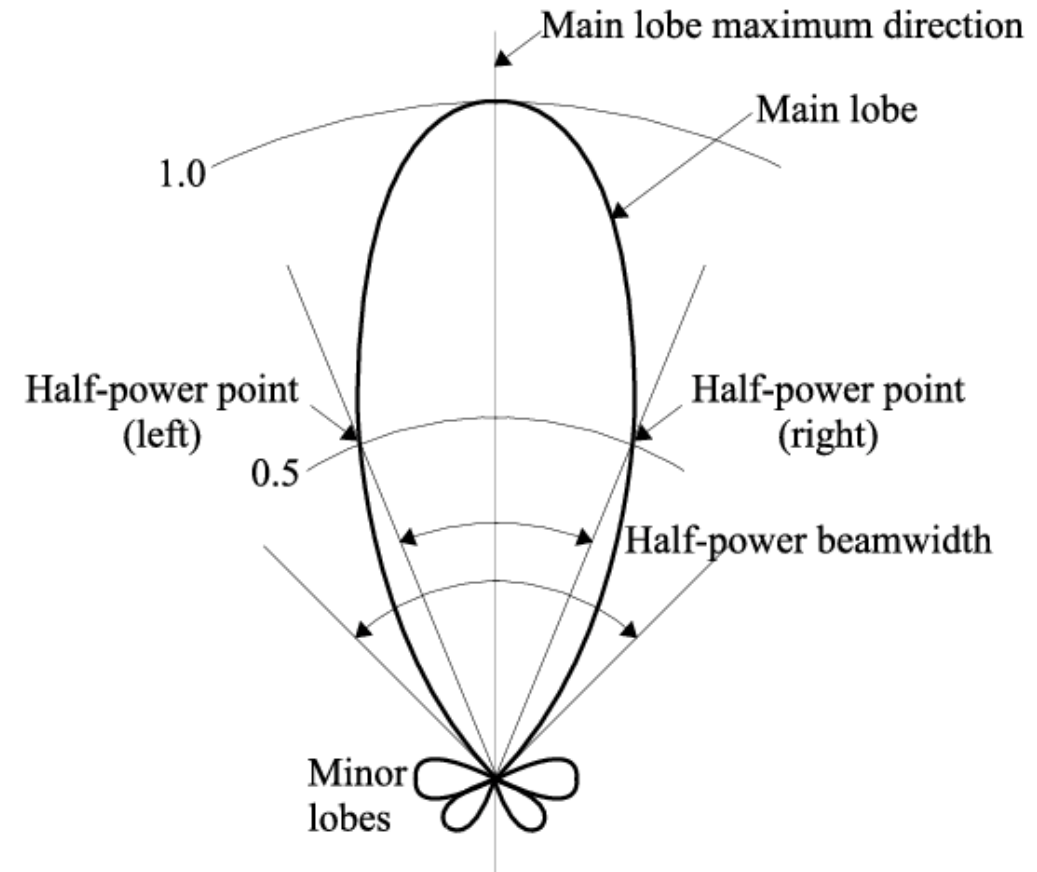
- **Gain:** Antenna gain refers to the ratio of the power at a particular point to the power with isotropic antenna, which gives a measure of power for the antenna.
- The gain of a real antenna is measured in:
  - **dB<sub>i</sub>** (decibels-isotropic) , if it compared with the isotropic antenna.
  - **dB<sub>d</sub>** (decibels-dipole), if it compared with half-wave dipole antenna
- When an isotropic antenna is compared with itself, the result is a gain of  $10\log_{10}(1)$  or 0 dB<sub>i</sub>.



# Basic Antenna Parameters

## Directivity

- **Directivity** is the maximum directive gain of an antenna.
- The directivity of an isotropic source is **unity** since its power is radiated equally well in all directions.
- For all other sources, the maximum directivity will always be greater than unity, and it is a relative "figure of merit" which gives an indication of the directional properties of the antenna as compared with those of anisotropic source.



# Basic Antenna Parameters

## Radiation resistance

- **Radiation resistance** represents how much of the power fed into an **antenna** is actually turned into useful radio waves (electromagnetic radiation) **instead of** being **wasted** as **heat** in the antenna.
- When you send power into an antenna, it does two things:
  - 1) **Radiates energy as electromagnetic waves (the useful part).**
  - 2) **Loses some energy as heat (wasted energy, due to resistance in the antenna's material).**
- The "**radiation resistance**" is like a special resistor that converts part of the power into radio signals instead of heat and given as:

$$R_r = \frac{P_{rad}}{I^2}$$

$P_{rad}$  is the total radiated power

$R_r$  is the radiation resistance.

$I$  is the current flowing through the antenna.

# Basic Antenna Parameters

## Radiation resistance

### Scenario: Calculating Radiation Resistance of an Antenna

Suppose you have a transmitting antenna that radiates **100 W** of power into free space when a **current of 2 A** flows through it. You are tasked with finding the **radiation resistance ( $R_r$ )** of the antenna.

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#### Step-by-Step Solution:

1. Given Data:

- Radiated power ( $P_{rad}$ ) = 100 W
- Current ( $I$ ) = 2 A

2. Formula for Radiation Resistance:

$$R_r = \frac{P_{rad}}{I^2}$$

3. Substitute the Values:

$$R_r = \frac{100}{(2)^2} = \frac{100}{4} = 25 \, \Omega$$

# Basic Antenna Parameters

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## Efficiency

- The efficiency of an antenna relates the **power delivered to the antenna** and **the power radiated or dissipated within the antenna**.
- A high efficiency antenna has most of the power present at the antenna's input radiated away.
- A low efficiency antenna has most of the power absorbed as losses within the antenna, or reflected away due **to impedance mismatch**.
- Antenna efficiency is given as:

$$e = \frac{P_{rad}}{P_{in}}$$

# Basic Antenna Parameters

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## Bandwidth

- Bandwidth of an antenna refers to the range of frequencies over which the antenna can operate correctly.
- Outside this range the levels of reactance rise to levels that may be too far for satisfactory operation

$$BW = F_H - F_L$$

where  $F_H$  is the highest frequency,  $F_L$  is the lowest.

- Antenna bandwidth is important where radio transmitters may damage if the antenna is operated outside its operating frequency range and the radio transmitter is not protected.
- Different types of antennas have different bandwidth limitations.

# Basic Antenna Parameters

## Beam Width

### Beam width:

The angle between the two directions in which the radiated power is half of the maximum value of the beam.

### Main lobe

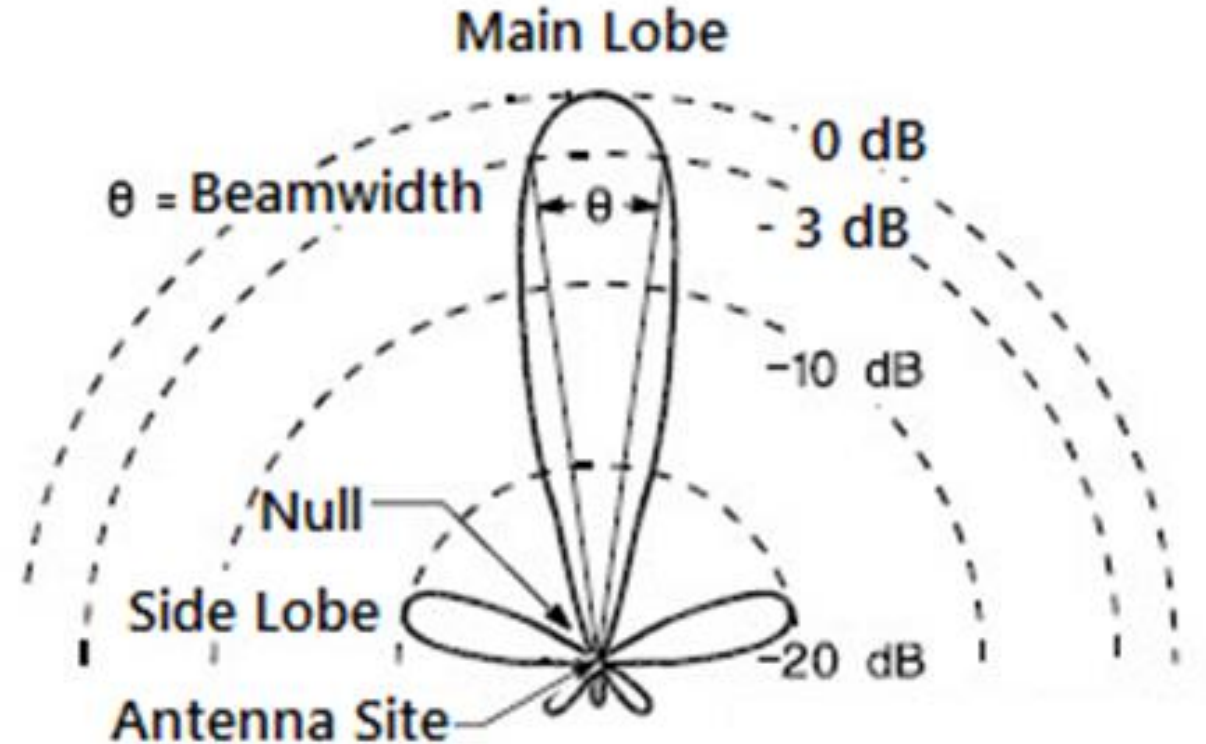
Lobe containing the maximum radiated power in the main direction.

### Side-lobes

Lobes containing small radiated power in other directions.

### Null

zone in which the effective radiated power is at a minimum and is useful for suppression of interfering signals in a given direction.



# Basic Antenna Parameters

## Front-to-Back Ratio

### Front:

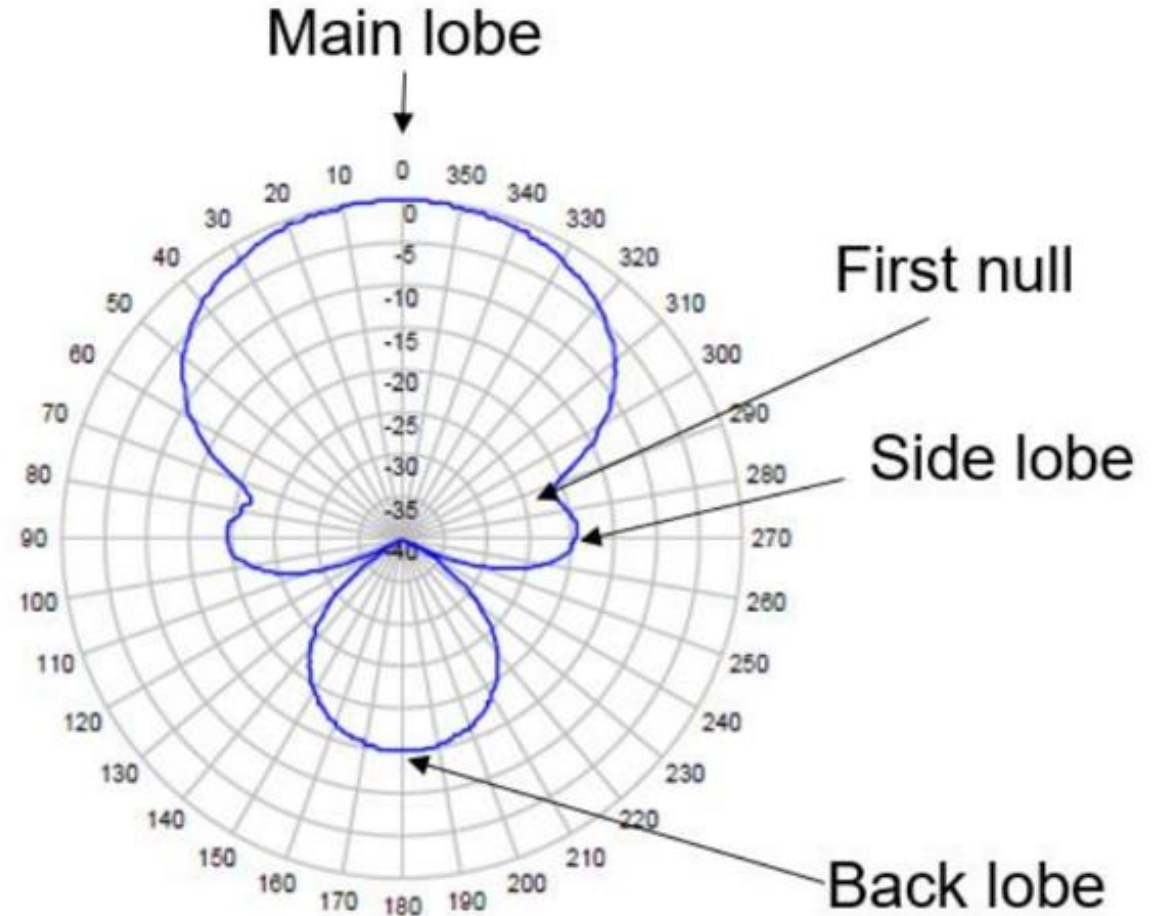
The direction of maximum radiation is in the horizontal plane is considered to be the front of the antenna.

### Back:

The back is the direction  $180^\circ$  from the front

### Major (Main) and Minor (Side) Lobes:

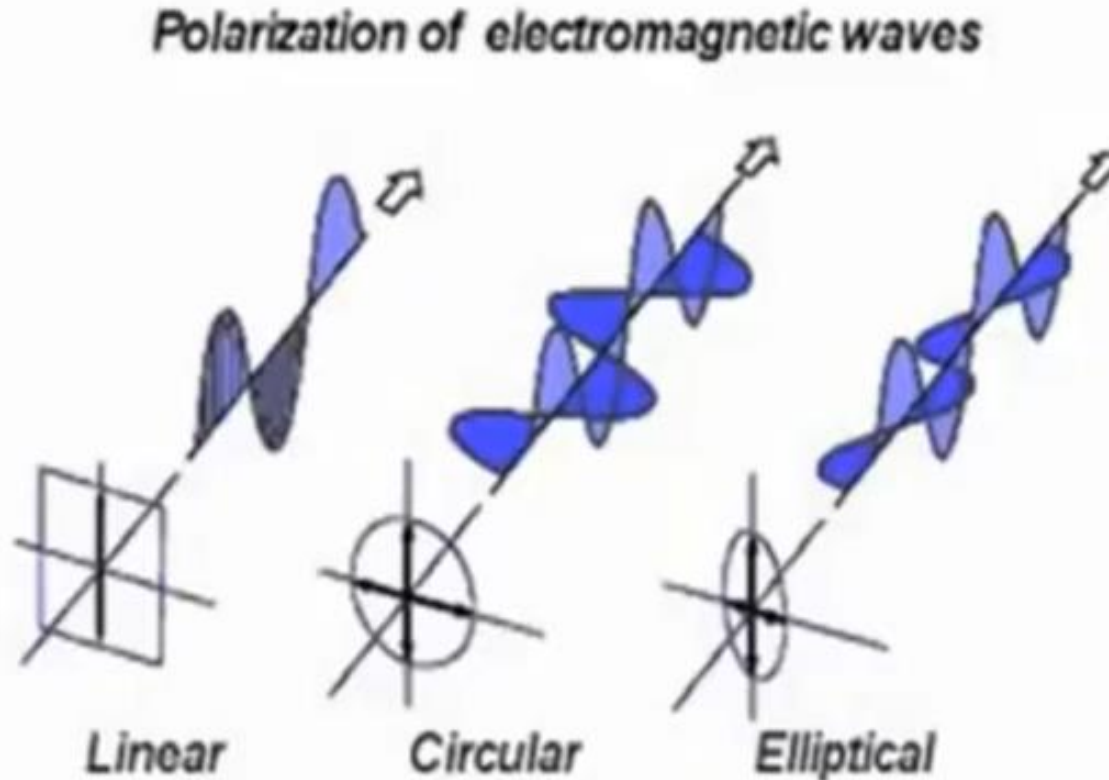
- The antenna has one major lobe and a number of minor ones
- Each of these lobes has a gain and a beam width which can be found using the diagram Azimuth



# Basic Antenna Parameters

## Polarization

- ❑ **Polarization** is the orientation of the **electric field** of an electromagnetic wave reference to the **ground plane**.
- ❑ Types of polarization are:
  - **linear polarization**
    - Vertical
    - Horizontal
  - **circular polarization**
    - Left hand
    - Right hand.
  - **Elliptical polarization**
    - Left hand
    - Right hand



# Antenna Applications

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- ❑ They are used in systems such as
- ❑ Radio broadcasting
- ❑ Broadcast television
- ❑ Two-way radio
- ❑ Communication receivers
- ❑ Radar
- ❑ Cell phones
- ❑ Satellite communications.

# Antenna Applications

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## **Example 1**

How much stronger a 17 dBi antenna receives (transmits) the signal compared to the isotropic antenna?

## **Solution**

Let

Power of isotropic antenna =  $P_{\text{iso}}$

Power of 17 dBi antenna =  $P$

We have

$$17 = 10\log_{10}(P/P_{\text{iso}})$$

Thus  $P/P_{\text{iso}} = 10^{1.7} = 50.12$ , i.e., the 17 dBi antenna will receive (transmit) the signal 50.12 times stronger than the isotropic antenna albeit using the same transmit power.

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# Note

- ❖ **Antennas** are designed to transmit or receive a specific frequency band.
- ❖ **For example**, antennas used in wireless routers operating with 2.5/5 GHz are too small to receive TV signals operating with 700 MHz.



Thank You